Exp6: Secure a Docker container by implementing best practices for user permissions and network security.

Aim: To secure a Docker container by:

- Running containers as a **non-root user**
- Restricting **network exposure**
- Applying basic **Docker security best practices**

Software Requirements

- Ubuntu 22.04 LTS
- Docker Engine
- Docker Compose (optional)

PART A: USER PERMISSION SECURITY (NON-ROOT CONTAINER)

Step 1: Create Project Directory

```
mkdir docker-secure-app
cd docker-secure-app
```

```
cse@cse-B760M-H:~$ mkdir docker-secure-app
cse@cse-B760M-H:~$ cd docker-secure-app
cse@cse-B760M-H:~/docker-secure-app$
```

Step 2: Create a Simple Python App

Create **app.py**:

```
nano app.py
```

Paste:

```
print("Secure Docker Container Running as Non-Root User")
```

```
  GNU nano 6.2
print("Secure Docker Container Running as Non-Root User")
```

Save and exit.

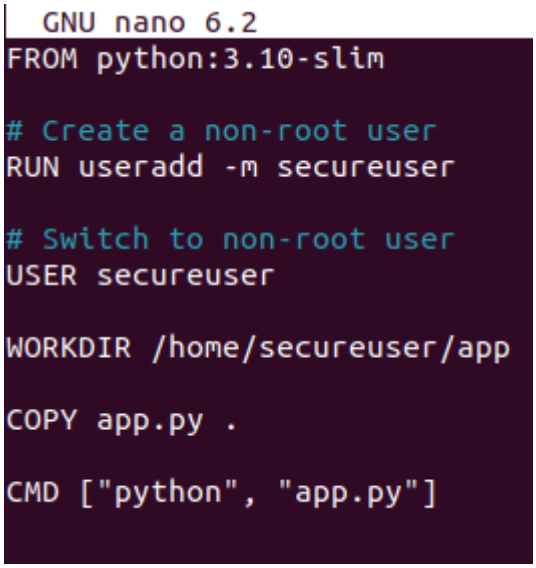
Step 3: Create a Secure Dockerfile

Create **Dockerfile:**

```
nano Dockerfile
```

Paste:

```
FROM python:3.10-slim


# Create a non-root user
RUN useradd -m secureuser


# Switch to non-root user
USER secureuser


WORKDIR /home/secureuser/app



COPY app.py .



CMD ["python", "app.py"]
```



Save and exit.

Step 4: Build the Secure Image

```
docker build -t secure-python-app .
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker build -t secure-python-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/6 : FROM python:3.10-slim
 ---> 988b1bc80158
Step 2/6 : RUN useradd -m secureuser
 ---> Running in 567cff0d28cc
 ---> Removed intermediate container 567cff0d28cc
 ---> 6e60df3772de
Step 3/6 : USER secureuser
 ---> Running in 077b855e148f
 ---> Removed intermediate container 077b855e148f
 ---> 0e6bff75b4b1
Step 4/6 : WORKDIR /home/secureuser/app
 ---> Running in 935e7729df17
 ---> Removed intermediate container 935e7729df17
 ---> f8b3c1b3d025
Step 5/6 : COPY app.py .
 ---> 955c45e1d4e4
Step 6/6 : CMD ["python", "app.py"]
 ---> Running in bcb926f5349a
 ---> Removed intermediate container bcb926f5349a
 ---> d1ea5417fde9
Successfully built d1ea5417fde9
Successfully tagged secure-python-app:latest
cse@cse-B760M-H:~/docker-secure-app$
```

docker images

```
cse@cse-B760M-H:~/docker-secure-app$ docker images
REPOSITORY                  TAG         IMAGE ID       CREATED          SIZE
secure-python-app           latest      d1ea5417fde9   32 seconds ago   122MB
docker-compose-web-db_web   latest      34f57e112a0b   4 hours ago      275MB
mysql                       8.0         b3197d59eb3b   14 hours ago     786MB
node-static-app             latest      b485ad226199   3 days ago       127MB
python-fileops              latest      fe055b321a21   4 days ago       122MB
simple-web-app              latest      5557786ca37d   7 days ago       134MB
hello-world-image           latest      b7e187b4b9ab   7 days ago       77.9MB
python                      3.10-slim   988b1bc80158   2 weeks ago      122MB
ubuntu                      22.04       65c77cbc27c2   6 weeks ago      77.9MB
node                        18-alpine   ee77c6cd7c18   10 months ago    127MB
cse@cse-B760M-H:~/docker-secure-app$
```

Step 5: Run the Container

```
docker run secure-python-app
```

Expected Output:

```
Secure Docker Container Running as Non-Root User
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker run secure-python-app
Secure Docker Container Running as Non-Root User
cse@cse-B760M-H:~/docker-secure-app$
```

Step 6: Verify User Inside Container

```
docker run -it secure-python-app /bin/bash
```

Inside container run:

```
whoami
```

Expected output:

```
secureuser
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker run -it secure-python-app /bin/bash
secureuser@7cee500a0def:~/app$ whoami
secureuser
secureuser@7cee500a0def:~/app$
```

Exit from the container:

```
exit
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker run -it secure-python-app /bin/bash
secureuser@7cee500a0def:~/app$ whoami
secureuser
secureuser@7cee500a0def:~/app$ exit
exit
cse@cse-B760M-H:~/docker-secure-app$
```

PART B: NETWORK SECURITY (PORT & NETWORK ISOLATION)

Step 7: Avoid Exposing Unnecessary Ports

Insecure:

Do not run this command

```
docker run -p 5000:5000 secure-python-app
```

Secure:

Instead run this command

```
docker run secure-python-app
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker run secure-python-app
Secure Docker Container Running as Non-Root User
cse@cse-B760M-H:~/docker-secure-app$
```

```
docker ps -a
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker ps -a
CONTAINER ID   IMAGE               COMMAND                  CREATED              STATUS                          PORTS     NAMES
e4561fb866ad   secure-python-app   "python app.py"          About a minute ago   Exited (0) About a minute ago             zealous_hypatia
7cee500a0def   secure-python-app   "/bin/bash"              8 minutes ago        Exited (0) 4 minutes ago                  tender_goldwasser
584ceb7ef261   secure-python-app   "python app.py"          9 minutes ago        Exited (0) 9 minutes ago                  romantic_wescoff
88d1a29c9665   node-static-app     "docker-entrypoint.s…"   3 days ago           Exited (137) 3 days ago                   pensive_blackburn
daefddb6f37b   python-fileops      "/bin/bash"              4 days ago           Exited (0) 4 days ago                     stoic_haibt
38274fb5d628   python-fileops      "python file_ops.py"     4 days ago           Exited (0) 4 days ago                     frosty_mayer
b116a6238394   simple-web-app      "python app.py"          7 days ago           Exited (137) 7 days ago                   competent_cray
72326bcd11bc   hello-world-image   "echo 'Hello World'"     7 days ago           Exited (0) 7 days ago                     eloquent_dirac
cse@cse-B760M-H:~/docker-secure-app$
```

Step 8: Create a Private Docker Network

```
docker network create secure-net
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker network create secure-net
981bd723aa8fba27e4d33f4d2ca62d621ea3012dc0f94567db384b9a870136cd
cse@cse-B760M-H:~/docker-secure-app$
```

```
docker network ls
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker network ls
NETWORK ID     NAME         DRIVER    SCOPE
d6fd0a757362   bridge       bridge    local
288dd6f562fb   host         host      local
37ff9eb72b27   none         null      local
981bd723aa8f   secure-net   bridge    local
cse@cse-B760M-H:~/docker-secure-app$
```

Step 9: Run Container in Isolated Network

```
docker run --network secure-net secure-python-app
```

```
cse@cse-B760M-H:~/docker-secure-app$ docker run --network secure-net secure-python-app
Secure Docker Container Running as Non-Root User
cse@cse-B760M-H:~/docker-secure-app$
```

PART C: ADDITIONAL SECURITY BEST PRACTICES

1. Use Minimal Base Images

```
FROM python:3.10-slim
```

Note: *Why?*

Smaller attack surface
Faster builds

2. Drop Linux Capabilities

```
docker run --cap-drop ALL secure-python-app
```

Note: *Containers don't need all Linux privileges*

3. Read-only File System

```
docker run --read-only secure-python-app
```

Note: *Prevents file tampering*

Result

The Docker container is secured by running it as a non-root user and applying network isolation and security best practices.